SECTION 1 – OPERATION

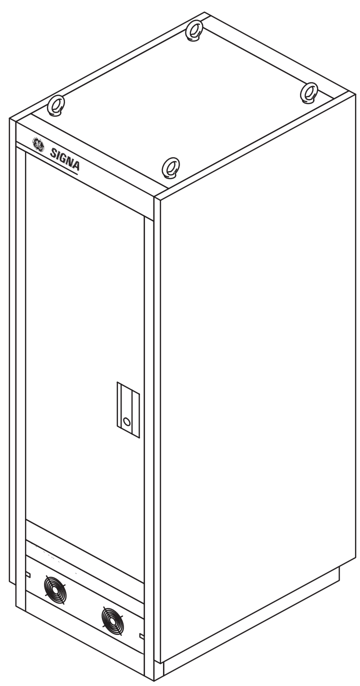
1–1 INTRODUCTION

The Compact Power Distribution Unit (PDU) is a three–phase power distribution unit. For output load distribution there is a 42–pole load center with secondary output breakers.

Compact PDU models 46–317099 P8 through P16 except P12 include a three–phase voltage regulator. Each phase is regulated independently. For an input voltage range between +4.5% surge to –13.5% sag about nominal rated voltage, Compact PDU typically regulates output line–to–line voltages to within 1.5% of nominal voltage.

Compact PDU models 46–317099 P8 through P16 contain a low–pass filter with high–energy surge suppression. Energy transients up to 12,400 Joules (line to line) or 6,200 Joules (line to neutral) are clamped and attenuated. Maximum line–to–line clamping voltage is 775 V. The operating frequency range is 47–63 Hz. The filter damping circuit reduces overshoot. Typical damping time is 500 microseconds.

A Compact PDU is shown in Illustration 1–1.



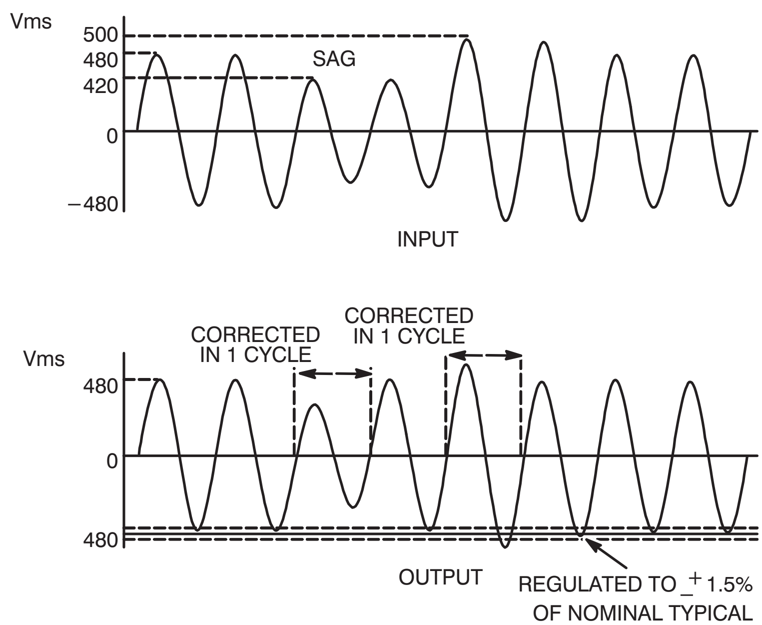
COMPACT PDU

ILLUSTRATION 1–1

1–1 INTRODUCTION (continued)

Models 46–317099P8 through P16 except P12 have tapped electronic voltage regulators. Six taps are on each primary winding of the transformer. Each tap is controlled by a silicon–controlled rectifier (SCR) switch and can become active at any time. Input line–to–line voltage and line current is monitored for each phase. Peak voltage is compared with a desired reference. Depending on this comparison, the regulation logic activates the proper SCR switch when the phase current becomes zero. This changes the primary–to–secondary turns ratio which corrects output voltage to the desired nominal. The correction process starts in less than one cycle. See Illustration 1–2.

The difference between the highest line–to–line voltage and the lowest line–to–line voltage on the output is within 2% of the nominal voltage.



REGULATED SYSTEM RESPONSE
ILLUSTRATION 1–2

Model 46–317099P12 is the same basic PDU as described above, but with manual voltage adjustment only. This model does not have the SCR regulation panels, control logic printed wiring board (PWB) assembly or the bypass switch. Six fixed taps on the transformer coils allow manual adjustment to compensate for site feeder voltage variations. Also three fixed taps on the transformer allow manual adjustments or 380, 415, or 480 VAC inputs.

1–2 BENEFITS

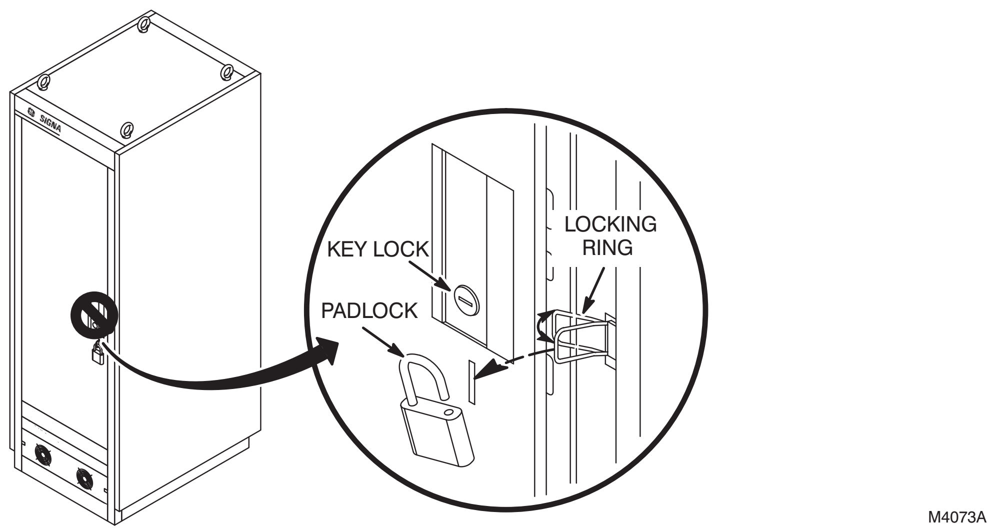
The regulated PDUs, Models P8–P16 except P12, protect electronic equipment from an overvoltage condition during line power surges and an undervoltage condition during sags. Models 46–317099P8 through P16 contain a filter which protects the medical equipment load by attenuating transients present on utility power lines. Overvoltage, undercurrent, and transient conditions stress electronic components and decrease system reliability.

Voltage fluctuations affect the image quality of imaging systems. Artifacts are introduced under sag conditions. Regulating power maintains image quality, resulting in more consistent images.

1–3 FRONT PANEL (A8)

The outer door is fitted with an opening for a lockout hasp attached to the frame. To see the front panel, open the outer door. See Illustration 1–3. To lockout access to Compact PDU:

1. Open outer door.
2. Swing out lockout hasp.
3. Close outer door so hasp extends through door opening.
4. Put padlock or other lockout device through hasp opening to lockout access to Compact PDU.
5. Tag padlock.



LOCKOUT DETAIL

ILLUSTRATION 1–3

The following switches and indicators are visible and available to the operator when the outer door is open. See Illustration 1–4.

1–3–1 Main Circuit Breaker (A1)

The main circuit breaker turns Compact PDU power on and off. It is shunt tripped by activation of thermal switch in system electronics, an SCR failure or blown fuselink, power off switch, hot switching of the bypass switch, and by activation of the thermal switch in the main transformer. The main circuit breaker will not close if bypass switch is in AUTO position and if SCR failure light is on. The main circuit breaker will not close if red LED on Accuvar of input filter is on.

1–3–2 Power Off (A3)

CAUTION

Facility power is still present at input side of main circuit breaker when tripped. To remove all power from the Compact PDU, turn facility circuit breaker OFF, lock, and tag.

DANGER!!

LETHAL VOLTAGE IS PRESENT AT FUSES F1, F8 AND TRANSFORMER T5 WHEN THE MAIN INPUT CIRCUIT BREAKER IS OFF.
TURN FACILITY CIRCUIT BREAKER OFF, LOCK AND TAG BEFORE SERVICING THE UNIT.

Press power off switch to trip main circuit breaker and turn power off to all loads.

1–3–3 Emer. Stop Reset (A5)

Note

In the Signa Advanatage System, The Gradient Amplifier Cabinet, Penetration Cabinet and RF Amplifier Cabinet load panel circuit breakers are tripped when:

- Emergency Stop is activated
- Activated from the PDU control board
- F1 or F8 opens
- 24 VDC supply fails.

After using an Emergency Stop switch, press Emer. Stop Reset prior to resetting tripped breakers.

Note

If facility power is lost or turned off, then on the return of power, before turning the Compact PDU main breaker on (if off), press Emer. Stop Reset.

1–3–4 SCR Failure Light (A15) (Models 46–317099 P8 through P16 except P12)

When an SCR malfunctions, SCR failure light is illuminated and main circuit breaker is tripped if bypass switch is in AUTO position.

1–3–5 Fail Reset Switch (A4) (Models 46–317099 P8 through P16 except P12

Press fail reset switch to clear SCR failure light when SCR malfunction has been corrected. The reset switch is used when bypass switch is in the AUTO position.

1–3–6 Bypass Switch (A2) (Models 46–317099 P8 through P16 except P12)

In AUTO position, line–to–neutral voltages are regulated. In MAN (manual) position, regulation logic is bypassed and input power is applied to load via transformer without regulation. Select the MAN position when there is a problem with regulation logic circuitry (typically indicated by an SCR failure light).

CAUTION

Turn off input power before switching bypass switch.

Rotating bypass switch with power applied trips the main circuit breaker. After rotating bypass switch, before resetting the main circuit breaker, press Emer. Stop Reset.

1–3–7 Test Points (TP1–TP7) (Models 46–317099 P8 through P16 except P12)

Test points allow monitoring of input line–to–line, output line–to–neutral, and output line–to–line voltages with standard field instruments. Test point voltage measurements are attenuated by a factor of 100 (e.g. 400 V = 4 V).

1–3–8 Load Panel (A5)

Below the front panel are circuit breakers which control power to loads.

1–3–9 Service Outlets (A12)

To access service outlets, open outer door and lower spring–loaded electrical outlet cover. See Illustration 1–4. Service outlets provide connections for the superconducting shim and magnet power supplies, and dual 120 V 15 A utility receptacles for tools such as a trouble light or scope. To connect to service outlets:

6. Unlock outer door and open.
7. Lower spring–loaded electrical outlet cover.
8. Plug connector into appropriate outlet.

With outer door open, outlet cover swings closed when cords are removed from service outlets. After use, close and lock outer door.

1–3–10 Output Status Lights (DS1–DS3)

Three lights on the front panel, labeled DS1, DS2 and DS3, indicate normal operation. The lighted condition indicates power is applied. If one or more lights are not illuminated, this indicates there is no output on that phase.

1–3–11 Fuses (F1, F8)

The two fuses on the back of the front panel (F1, F8) for models P8–P16 except P12 protect the T5 primary (A9T5), which supplies power to the SCR sense board. See illustration 3–18 for location. On model 46–317099P12, F1 and F8 are on the load center bracket assembly. See illustration 3–23 for location.

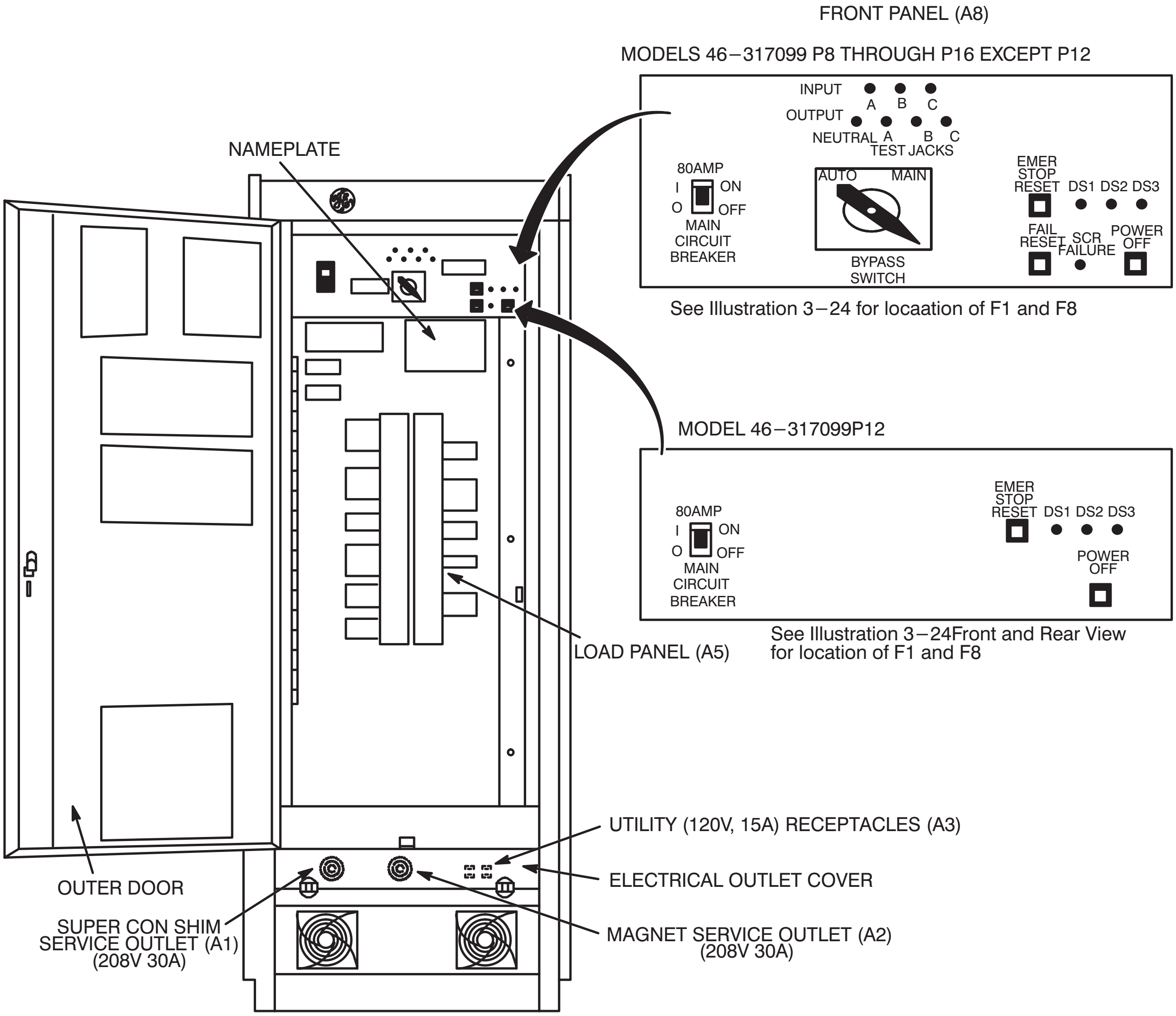
1−3−11 Fuses (F1, F8) (continued)

When either F1 or F8 opens, the gradient amplifier cabinet, penetration cabinet, RF amplifier cabinet and optional load panel circuit breakers with shunt trips are tripped.

DANGER!!

LETHAL VOLTAGE IS PRESENT AT FUSES F1, F8 AND TRANSFORMER T5 WHEN THE MAIN INPUT CIRCUIT BREAKER IS OFF.
TURN FACILITY CIRCUIT BREAKER OFF, LOCK AND TAG. BEFORE SERVICING THE UNIT.

1–3–11 Fuses (F1, F8) (continued)



FRONT PANEL AND LOAD PANEL

ILLUSTRATION 1–4

1–4 STARTUP

After installing Compact PDU or after moving trailer containing medical system, start system with the following procedure:

1–4–1 Tools Needed

- Phase meter
- Voltmeter

1–4–2 Startup Procedure

If Compact PDU fails any of the following checks, refer to Section 3–3, TROUBLESHOOTING.

1. Open outer door of Compact PDU.
2. Turn all load panel circuit breakers OFF.
3. Turn main circuit breaker OFF.
4. Turn facility circuit breaker OFF, lock, and tag.
5. Measure input voltage at test points on front panel, for 0 VAC before proceding. If not zero recheck steps 3 and 4. On models P12, measure for voltage at filter terminals L1, L2, L3 located on right front of the filter assembly. See illustration 3–10.
6. Loosen load panel door with a 90° turn of two assembly screws on right. Remove 1/4–20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
7. Check that all accessible power connections are tight: phase A, B, C, neutral, and ground.
8. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Install 1/4–20 screw in center of right edge of door.
9. Unlock facility circuit breaker and turn ON
   OR
   Connect power to trailer.

DANGER!!

FATAL SHOCK HAZARD!! LETHAL VOLTAGES EXIST WITHIN COMPACT PDU DURING THIS CHECK. FOLLOW THE STEPS BELOW EXACTLY. FAILURE TO DO SO COULD RESULT IN SEVERE INJURY OR DEATH.

10. Check that input power is properly phased at connection to utility. Use a phase meter.

1−4−2 Startup Procedure (continued)

CAUTION

Be sure input power is properly phased. Damage to loads may result if input power is not phased A, B, C (clockwise phase rotation).

11. Verify that there is no obstruction on top of Compact PDU which restricts air flow.
12. Measure voltages at L1, L2, L3 of TB10 (A2). Verify that voltage is within limits of input voltage range. Refer to Table 2−3.
13. Configure the transformer taps. Refer to Section 3−2, CONFIGURATION.
14. Press Emer. Stop Reset switch.
15. Turn main circuit breaker ON.
16. Models 46−317099 P8 through P16 except P12: Measure facility line−to−line voltage at front panel input test points and compare with model voltage specification on nameplate located over load panel inside outer door.

Note

There is a 100:1 voltage step−down at front panel test points. For example, the measure of 480 VAC input voltage is 4.8 VAC at front panel test points.

17. Turn main circuit breaker OFF.
18. Models 46−317099 P8 through P16 except P12: Turn bypass switch to AUTO position.
19. Turn main circuit breaker ON.
20. Press power off switch. Main circuit breaker will trip.
21. Reset and turn main circuit breaker ON.
22. Verify that three output power status lights on front panel are lit (6 seconds delay).
23. Models 46−317099 P8 through P16 except P12: Verify output line−to−neutral voltage at test points of 1.20 $\pm$ 0.03 V. If not, refer to Section 3−1−4, Calibrate Logic Board.
24. Press emer. stop reset switch.
25. Turn load panel circuit breakers ON.
26. Check that all loads are operational.
27. Close outer door.

1–5 OPERATION

The Compact PDU is designed for unattended continuous operation.

1–5–1 Normal Condition

The normal condition of Compact PDU is as follows (See Illustration 1–4 for location):

1. Main circuit breaker ON.
2. Models 46–317099 P8 through P16 except P12: Bypass switch in AUTO position.
3. Load panel circuit breakers ON.
4. Output indicator lights DS1, DS2, DS3 lit.

Note

If main circuit breaker is tripped use the following restart procedure:

A. Locate and correct condition which caused main circuit breaker to trip.
B. Reset main circuit breaker.

5. All Models: Emer. stop reset switch extinguished.

Note

If any load panel circuit breaker is intentionally tripped remotely, emer. stop reset switch must be pressed to reconnect the circuit and breaker must be manually reset at load panel after fault is corrected. The trippable circuit breakers (gradient, RF, penetration cabinet and some optional breakers) will also trip if F1 or F8 opens.

Note

If a load panel circuit breaker is tripped use the following restart procedure:

A. Locate and correct condition which caused load panel circuit breaker to trip.
B. Reset appropriate load panel circuit breaker.

6. Models 46–317099 P8 through P16 except P12: SCR failure light extinguished.

1–5–2 Special Condition

On models 46–317099 P8 through P16 except P12, the SCR failure light is lit if an SCR fails or if a fuselink opens. If an SCR fails or a fuselink opens, use the following procedure to continue operating load:

Note

The following procedure is only for temporary operation. Voltage surge and sag protection normally available in Models 46–317099 P8 through P16 except P12 is not available for *any* load while the bypass switch is in MAN position. Refer to Section 3–3–4, SCR troubleshooting for troubleshooting SCR failures.

1. Turn main circuit breaker OFF.
2. Turn bypass switch from AUTO to MAN position. Then turn main circuit breaker ON and continue operation.